A Comparison of Neural Architectures for Movie Poster Genre Classification

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Neural architectures

- Krizhevsky et al. (2012) won the 2012 ImageNet competition with AlexNet, the network that made deep convolutional models the default for image classification.
- Simonyan and Zisserman (2015) built VGG16 from a single repeated 3×3 convolution, on the argument that depth buys expressiveness more cheaply than width.
- He et al. (2016) have each block of a residual network (ResNet) learn a residual against its own input, so the gradient has a path that skips the block entirely.
- Jocher et al. (2023) replace the detection head of the real-time object detector You Only Look Once (YOLO) by a whole-image classifier in the YOLOv8 classification branch.

Zero-shot prediction and efficient backbones

- Radford et al. (2021) trained CLIP to align images and text in a shared embedding space, so a classification task can be posed without any further training.
- Tan and Le (2021) add progressive learning in EfficientNetV2, which begins training at a small image size under weak regularization and raises both as training proceeds.
- Vasu et al. (2023) designed Fast-ViT, a hybrid that keeps convolutions wherever a transformer would be expensive.

Research question

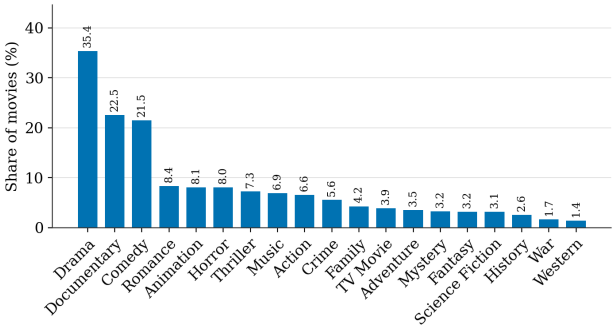
Question

How far does the genre signal of a movie poster survive as a machine-readable one across neural architectures, and at what cost?

Finding

1. We believe a movie poster carries enough genre signal to be worth modeling but not enough for a reliable label.
2. ResNets generally achieve the best results, reaching 54.5% accuracy on 19 genres and 69.7% on 3 genres.
3. A release decade indicator raises the accuracy of the network trained from scratch by about 4%, but transfer learning outperforms it in a fraction of the time.
4. VGG16 draws on the whole poster rather than its center and tolerates rotation.

Data



Distribution of genres.

- 557,539 posters in the Letterboxd collection on Kaggle (Simon, 2024)

→ 552,531 of the dominant shape $(345, 230, 3)$

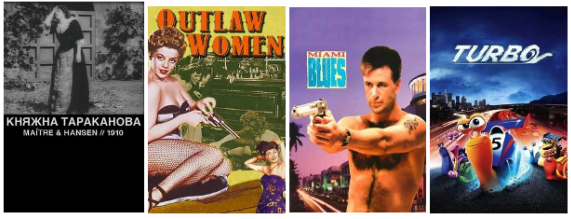
→ 548,835 with at most four genres

→ 424,102 for the CNN, released after 1970 with at most three genres

- Each movie carries one to four of 19 genres, led by Drama, Documentary and Comedy.
- The CNN keeps posters released after 1970, since the earlier material is sparse and stylistically distinct from the rest.

Convolutional neural network

- A 12-layer convolutional neural network (CNN) built from scratch, with 32 filters in its first 9 layers and 64 in the last 3, fixes the floor every later model is measured against.
- The sample contains 9,000 posters balanced across genres, 7,200 for fitting and 1,800 for testing, on three sets of 3 genres for releases after 1970 and after 2000.
- Poster design is not stationary, so the release decade is supplied as an indicator. Across repeated runs, the decade indicators raise accuracy by about 4%, yet the CNN barely performs adequately on 3 genres (Appendix B.1).



Poster design across four decades, from the 1910s (left) to the 2010s (right).

VGG16

Layer	Output shape	Parameters
Input	$(345, 230, 3)$	0
VGG16	$(10, 7, 512)$	14,714,688
Flatten	(35,840)	0
Dense, sigmoid	(128)	4,587,648
Dropout at 0.25	(128)	0
Output, softmax	(3)	387

Layers of the VGG16 classifier with output shapes and parameter counts.

- VGG16 keeps frozen ImageNet weights, so only the classifier on top is fitted.
- Adam minimizes categorical cross-entropy at a learning rate of 0.001.
- Training runs for 50 epochs in mini-batches of 32.

- On held-out posters of 3 genres, accuracy and F1 sit near 61%, and the area under the receiver operating characteristic curve (AUROC) reaches 79.0%.
- This is a clear advance over the CNN at a fraction of the training cost.

Confusion matrices under three transformations

Transformation	True genre	Action	Romance	Thriller
Monochrome, F1 = 59.4 %	Action	482 → 409	134 → 165	184 → 226
	Romance	119 → 80	510 → 507	150 → 192
	Thriller	169 → 133	170 → 177	482 → 511
Border removed, F1 = 45.7 %	Action	482 → 202	134 → 60	184 → 538
	Romance	119 → 63	510 → 244	150 → 472
	Thriller	169 → 45	170 → 40	482 → 736
Rotated, F1 = 54.9 %	Action	482 → 526	134 → 94	184 → 180
	Romance	119 → 247	510 → 345	150 → 187
	Thriller	169 → 252	170 → 115	482 → 454

Rows true, columns predicted genre; original count, then transformed count. The original F1 is 61.4%.

- Monochrome leaves the confusion matrix almost intact, so we infer that the network uses color chiefly to locate edges and that the grayscale gradient preserves them.
- Removing the border does considerable damage, so the classifier is not concentrating on the center in the way a human viewer does. Rotation costs little.

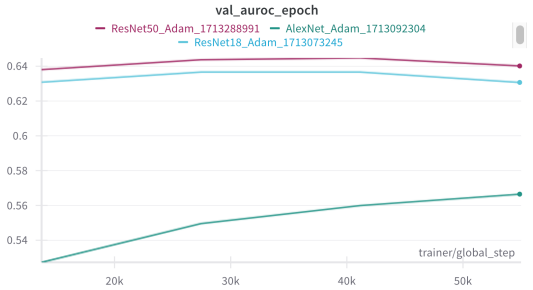
AlexNet and residual networks

Residual networks have each block learn a residual against its own input (He et al., 2016),

$$A^{(\ell)} = \varphi\Big(\mathcal{F}\Big(A^{(\ell-1)}; \theta^{(\ell)}\Big) + A^{(\ell-1)}\Big),$$

- $A^{(\ell-1)}$ is the input of block ℓ, $\mathcal{F}(\cdot\,; \theta^{(\ell)})$ the stack of convolutions inside it with parameters $\theta^{(\ell)}$ and φ the activation.
- ResNet-18 stacks basic blocks of two 3×3 convolutions and ResNet-50 bottleneck blocks of 1×1, 3×3 and 1×1 convolutions.
- If depth is what matters the two depths should separate, and if the residual connection is what matters both should stand clear of AlexNet.
- All three are fitted for 5 epochs with Adam and binary cross-entropy, with a sigmoid output since a movie may carry several genres.

Validation AUROC by epoch under Adam



Validation AUROC by epoch for AlexNet and ResNets under Adam.

- ResNet-50 is far deeper than ResNet-18, yet their training and validation curves nearly coincide, and ResNet-50 leads by a slim margin on every genre (Appendix A.3).
- ResNet outperforms AlexNet throughout, and this gap is therefore attributable to the residual connection rather than to depth alone.

YOLOv8

- YOLOv8 is included to test whether a detection backbone, which must localize objects rather than summarize a scene, reads a poster differently from a classification backbone.
- Two approaches file each poster under a folder named for its label.
- Multi-label A poster is copied into one folder per genre it carries.
- Multi-class A poster appears once, under a single genre.
- Both approaches train for 10 epochs with Adam. The multi-label approach barely learns, and filing one poster under several conflicting labels appears to be the cause.

CLIP

A poster X is classified without any further training by writing each of the G genres as a sentence T_g and returning the genre whose sentence lies closest to the poster,

$$\hat{y}(X) = \underset{g \in \{1,\dots,G\}}{\arg\max} \frac{f_I(X)^\top f_T(T_g)}{\|f_I(X)\| \, \|f_T(T_g)\|},$$

- f_I and f_T are the image and text encoders of CLIP, pre-trained contrastively on matched image and caption pairs (Appendix C.1).
- Image encoders ResNet-50, ResNet-50x4 at roughly four times the arithmetic, and the base vision transformer at patch sizes 32 and 16, ViT-B/32 and ViT-B/16.
- Text encoder Held fixed across the four, with the prompt *A movie poster of* {} *genre*.
- CLIP has never seen the poster dataset, so it gives the accuracy available with no task-specific training, which a fine-tuned model has to beat to justify its cost.

Fine-tuning

EfficientNetV2 and Fast-ViT keep the pre-trained body f_{body} frozen and replace the head that outputs 1,000 ImageNet classes with one that outputs the G genres. For EfficientNetV2, it is

$$\boldsymbol{\eta} = U_2 \operatorname{Drop}_{0.2}(U_1 \operatorname{Drop}_{0.2}(\mathbf{z}) + \mathbf{c}_1) + \mathbf{c}_2, \qquad \Pr(y = g \mid X) = \frac{\exp(\eta_g)}{\sum_{g'=1}^{G} \exp(\eta_{g'})},$$

- $\mathbf{z} = \mathcal{A}(f_{\text{body}}(X)) \in \mathbb{R}^q$ is the globally averaged feature vector, $q = 1{,}280$ for EfficientNetV2 and $q = 1{,}024$ for Fast-ViT.
- The two affine maps are separated by dropout, with U_1 of width 1,000. For Fast-ViT, the head is a single affine map to G outputs.
- Training minimizes cross-entropy using Adam for 10 or 22 epochs.

Time embedding

The time embedding variant adds the release year t, so that the head receives

$$\left(\mathbf{z}^\top, \mathbf{u}(t)^\top\right)^\top \quad \text{in place of} \quad \mathbf{z}, \qquad \mathbf{u}(t) \in \mathbb{R}^{4K},$$

- t enters as a one-hot vector over the K distinct release years, each coordinate of which indexes a learned four-dimensional embedding.
- The rows used for evaluation are the rows the models were fine-tuned on.

Accuracy after fine-tuning over 19 genres

Model	Epochs	Without time embedding	With time embedding
EfficientNetV2	10	45.6	44.1
EfficientNetV2	22	46.2	30.4
Fast-ViT T12	10	36.9	
Fast-ViT T12	22	39.8	
CLIP, zero-shot		32.9	

Accuracy after fine-tuning over 19 genres, against the best zero-shot encoder (%).

- The time embedding hurts rather than helps EfficientNetV2 at both epoch counts, so we did not carry it over to Fast-ViT.
- EfficientNetV2 leads Fast-ViT, and both lead the zero-shot reference, which is what fine-tuning is supposed to deliver.

Accuracy after fine-tuning over 3 genres

Model	Epochs	Accuracy	Training and inference time
EfficientNetV2	22	59.2	135 minutes
Fast-ViT T12	22	60.8	100 minutes
CLIP, zero-shot		60.9	20 minutes

Accuracy after fine-tuning over Comedy, Documentary and Drama, against the best zero-shot encoder (%).

- On Comedy, Documentary and Drama, zero-shot CLIP edges ahead of both fine-tuned models in 20 minutes against 100 and 135.
- This suggests that on an easy label set the textual description of the genre is worth more than task-specific fitting.

Evaluation metrics on 3-genre classification

Model	AUROC	F1	Precision	Accuracy
CNN, average	60.9	44.6	44.7	45.2
VGG16	79.0	61.4	62.0	61.4
AlexNet	71.6	61.5	62.8	61.8
ResNet-18	77.4	69.8	69.7	69.7
ResNet-50	77.5	69.7	70.2	69.7
YOLOv8, multi-label	60.4	39.6	39.6	39.6
YOLOv8, multi-class	53.8	39.6	39.6	39.6
CLIP, ResNet-50	74.4	52.8	55.8	53.2
CLIP, ResNet-50x4	71.6	45.4	51.2	51.0
CLIP, ViT-B/32	74.3	51.3	56.3	55.1
CLIP, ViT-B/16	79.8	60.0	60.5	60.9

Evaluation metrics on 3-genre single-label classification (%). CNN: mean of Appendix B.1; VGG16 to YOLOv8: Action, Romance, Thriller; CLIP: Comedy, Documentary, Drama.

- ResNets generally achieve the best results.
- CLIP with ViT-B/16 leads on AUROC.
- Training a CNN from scratch is not worth the compute.

Evaluation metrics on 19-genre classification

Model	AUROC	F1	Precision	Accuracy
AlexNet	54.5	13.4	22.7	42.9
ResNet-18	61.6	28.1	34.0	53.6
ResNet-50	62.6	29.9	33.6	54.5
YOLOv8, multi-label	82.8	29.7	29.7	29.7
YOLOv8, multi-class	82.0	29.0	29.0	29.0
CLIP, ResNet-50	75.9	16.7	19.3	25.5
CLIP, ResNet-50x4	78.2	18.4	20.4	25.5
CLIP, ViT-B/32	79.3	19.0	20.4	31.7
CLIP, ViT-B/16	81.3	21.0	22.0	32.9

Evaluation metrics on 19-genre single-label classification (%).

- ResNets deliver the strongest results, with ResNet-50 reaching 54.5% accuracy.
- YOLOv8 leads on AUROC.

Conclusion

- We compare a CNN built from scratch, four transferred ImageNet backbones, YOLOv8, zero-shot CLIP and two fine-tuned efficient backbones on 3 and 19 genres.
- ResNets generally achieve the best results, with 54.5% accuracy on 19 genres and 69.7% on 3. Training a CNN from scratch is not worth the compute.
- VGG16 is rotation tolerant and largely indifferent to color, but reliant on the whole image rather than its center.
- We believe a movie poster carries enough genre signal to be worth modeling but not enough for a reliable label.

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Code and replication data are available at *GitHub*.

Appendix A.3. AUROC by genre

Genre	AlexNet	ResNet-18	ResNet-50
Action	77.3	83.2	84.3
Adventure	72.1	77.1	78.6
Animation	87.9	94.5	95.2
Comedy	72.6	78.0	78.4
Crime	69.1	75.7	77.8
Documentary	75.9	83.4	83.7
Drama	66.6	74.6	75.0
Family	75.6	80.3	82.3
Fantasy	66.4	72.3	74.9
History	65.2	74.8	76.8

Genre	AlexNet	ResNet-18	ResNet-50
Horror	81.6	87.0	88.0
Music	67.6	78.2	80.0
Mystery	66.0	72.4	73.9
Romance	72.6	78.9	80.0
Science Fiction	72.5	78.6	81.0
TV Movie	70.0	78.2	80.1
Thriller	74.9	80.2	81.4
War	70.9	79.7	82.3
Western	83.8	92.1	93.3

AUROC by genre on the raw scores of the test half (%). The 19-genre table scores the hard prediction.

ResNet-50 leads ResNet-18 by a slim margin on every genre.

Appendix B.1. Evaluation metrics for CNN

Setting	AUROC	F1	Precision	Accuracy
Action, Drama, Romance, after 1970	60.2	42.6	42.9	44.5
Action, Drama, Romance, after 2000	66.2	46.6	45.4	44.9
Comedy, Drama, Romance, after 1970	62.0	45.6	46.3	47.3
Comedy, Drama, Romance, after 2000	62.1	45.6	45.6	46.5
Drama, Horror, Music, after 1970	62.5	45.2	45.3	46.6
Drama, Horror, Music, after 2000	53.0	41.7	42.4	41.4
Average	60.9	44.6	44.7	45.2

Evaluation metrics for CNN (%).

- Across repeated runs, the decade indicators raise accuracy by about 4%, which we take as confirmation that release period carries information the pixels do not supply on their own.
- Nevertheless, the CNN barely performs adequately on 3 genres.

Appendix B.2. VGG16 F1 by region and language

	Africa	Asia	Europe	North America	South America	Oceania
F1	54.7	61.3	60.3	64.8	57.6	59.2

VGG16 F1 by production region (%).

	English	Spanish	Cantonese	Hindi	Korean	Thai
F1	61.5	62.1	61.4	57.0	65.8	61.6

VGG16 F1 by spoken language (%).

- African and South American releases have the lowest F1 and also the fewest examples.
- Across languages, the spread is narrower, and Hindi is the only clear shortfall, while Korean releases score above the English baseline.

Appendix B.3. Zero-shot accuracy of CLIP by release period

Image encoder	All periods	After 1970	After 2000
ResNet-50	25.5	25.7	26.4
ViT-B/32	31.7	33.6	34.4

Zero-shot accuracy of CLIP over 19 genres by release period (%).

- Accuracy improves slightly as the window narrows.
- This indicates a real but small time-varying element in the data.

Appendix C.1. Contrastive pre-training of CLIP

For a batch of N matched pairs, the cosine similarity between the ith image X_i and the jth caption T_j is

$$s_{ij} = \frac{f_I(X_i)^\top f_T(T_j)}{\|f_I(X_i)\| \, \|f_T(T_j)\|},$$

and pre-training minimizes

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{2N} \sum_{i=1}^{N} \left[\log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N} \exp(s_{ij}/\tau)} + \log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{N} \exp(s_{ji}/\tau)} \right],$$

- f_I and f_T are the image and text encoders, and $\tau > 0$ is a learned temperature.
- Pre-training maximizes the similarity of the N matched pairs while minimizing it across the $N^2 - N$ mismatched ones.